

AI-Powered Interview Preparation Platform

Gemini AI-integrated platform for dynamic role-specific interview practice

[Tech Stack](#) | [React.js](#) · [Next.js](#) · [Gemini AI API](#) · [Node.js](#) · [REST APIs](#) · [SSR](#)

AI/LLM Integration	Next.js SSR	Product	Analytics	Real Users
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Project Overview

- A fully functional AI-driven interview preparation platform that integrates Google's Gemini AI API to generate dynamic, role-specific interview questions and deliver structured, actionable performance feedback. Built with Next.js for server-side rendering, it tracks per-session analytics and reached 20+ active users within 2 weeks purely through peer referrals.

Key Technical Highlights

- Gemini AI API integration for role-specific question generation — dynamically adapts to job title and domain (SWE, data, PM, etc.).
- Structured performance feedback engine that scores answers, identifies weak areas, and surfaces improvement suggestions per session.
- Rate-limit handling and graceful fallback logic to maintain platform stability when API quotas are approached.
- Next.js SSR frontend for fast initial page loads and SEO-friendly rendering of session history pages.
- Per-session analytics dashboard tracking accuracy trends, topic coverage, and weak-area heatmaps across multiple practice sessions.
- Reached 20+ active users within 2 weeks with zero marketing spend — growth entirely through peer referrals.

Architecture & Design Decisions

- Next.js SSR pages for session views; client-side React components for interactive Q&A; flow.
- Node.js backend orchestrates Gemini API calls, appending role context and difficulty tier to each prompt.
- Rate-limit middleware monitors API usage per user session; fallback serves cached question sets if quota is hit.
- Session data (questions, answers, scores) persisted via REST API to backend storage for analytics computation.

- Analytics engine aggregates per-session scores to compute accuracy trends and topic-level performance breakdowns.

Outcomes & Impact

- 20+ active users onboarded within 2 weeks through peer referrals — validated real product-market fit.
- Dynamic AI question generation eliminated the need for a static question bank, making the platform infinitely scalable in content.
- Structured feedback loop shown to improve user self-reported confidence in mock interviews after 3+ sessions.
- Demonstrated practical LLM integration skills: prompt engineering, API orchestration, rate limiting, and fallback design.